Group 2 – Explainable Potato Leaf Disease Detection using ViT & XAI

Objective:

The goal of this assignment is to develop an explainable deep learning framework for potato leaf disease detection using image data. Students will implement a Vision Transformer (ViT) model to classify potato leaf diseases and apply Explainable Artificial Intelligence (XAI) techniques to interpret model predictions and identify disease-affected regions.

Task Breakdown:

1. Dataset Collection & Preprocessing:

- Download a publicly available potato leaf disease dataset (e.g., PlantVillage).
- Select relevant classes: Early Blight, Late Blight, Healthy.
- Resize all images to a uniform size (e.g., 224×224).
- Apply preprocessing: normalization and data augmentation (rotation, blur, brightness, noise).
- Split dataset into Training (70%), Validation (20%), Testing (10%).

2. Labeling Strategy:

- Use predefined labels from dataset: Early Blight, Late Blight, Healthy.
- Ensure correct mapping between images and labels.

3. Dataset Preparation:

- Organize dataset into train/validation/test folders.
- Ensure balanced class distribution.
- Handle class imbalance using augmentation or class weights.

4. Model Implementation:

- Implement Vision Transformer (ViT).
- Use pretrained models (HuggingFace/PyTorch).
- Train using PyTorch or TensorFlow.
- Tune hyperparameters such as learning rate, batch size, and epochs.

5. Explainability (XAI):

- Apply Grad-CAM and Attention Rollout.
- Generate heatmaps showing disease regions.
- Interpret model behavior.

6. Evaluation & Analysis:

- Evaluate using Accuracy, Precision, Recall, F1-score.
- Plot confusion matrix and training curves.
- Visualize predictions.

7. Comparative Study:

- Compare ViT with CNN models (ResNet/VGG/MobileNet).
- Analyze performance and feature learning differences.

8. Real-World Simulation:

- Apply augmentations such as noise, blur, lighting variation.
- Evaluate robustness.

9. Final Report & Observations:

- Document methodology, results, and observations.
- Discuss strengths, limitations, and future scope.

Deliverables:

- Code Implementation
- Public GitHub Repository (mandatory)
- Report (PDF)
- Visualization outputs (heatmaps, graphs)

Discussion:

Discuss whether the Vision Transformer captures global disease patterns better than CNNs.
Analyze if XAI helps in identifying meaningful disease regions.
Reflect on the role of explainability in real-world agricultural applications.